

Introduction to the Course 'Quantum Computing with Qiskit and Advanced Algorithms'

Welcome to the course **Quantum Computing with Qiskit and Advanced Algorithms**. The third part of the 3 course Specialization in [The Complete Quantum Computing Course for Beginners](#) .

The course is structured into seven comprehensive modules:

1. **Qiskit 101:** In this module, we will introduce the fundamental concepts of Qiskit, including classical and quantum gates, and build your first quantum circuit. You will also learn how to simulate your circuits and run them on a real quantum computer using IBM's quantum platform.
2. **Teleportation:** In this module, we will explore the mechanics of quantum teleportation, including phase, the Bloch sphere, and superdense coding. You will also implement teleportation protocols directly in Qiskit.
3. **Bernstein Vazirani:** In this module, we will dive deep into the Bernstein Vazirani algorithm, exploring its quantum advantages over classical approaches. You will also learn how to improve and apply this algorithm within Qiskit.
4. **Deutsch:** In this module, we will study the Deutsch algorithm, one of the foundational quantum algorithms, and implement it within Qiskit. You will gain hands-on experience in developing and testing this algorithm on quantum hardware.
5. **Grover's:** In this module, we will introduce Grover's search algorithm, contrasting it with classical search methods. You will learn to implement Grover's algorithm in Qiskit and apply it to a practical scenario involving optimization.
6. **Shor's:** In this module, we will explore Shor's algorithm and its profound impact on cryptography. You will also delve into key components such as the Quantum Fourier Transform and Quantum Phase Estimation, and implement the algorithm within Qiskit.
7. **Next Steps:** In this module, we will look at the next steps in your quantum computing journey, focusing on Qiskit documentation and quantum hardware. You will also explore resources for continuing your learning and expanding your quantum skills.

Target Learner: This course is aimed at professionals and students with a foundational knowledge of linear algebra and classical computing. Experience with Python is recommended, as Qiskit relies heavily on Python programming.

Learning Objectives:

1. Learn to use Qiskit for implementing and running quantum algorithms.
2. Master advanced quantum algorithms such as teleportation, Bernstein-Vazirani, Deutsch, Grover's, and Shor's algorithms.
3. Apply quantum computing concepts in practical scenarios using Qiskit.